

Machine Learning Based Classification of QUIC Network Traffic for Encrypted Traffic Analysis

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Abstract—With the widespread adoption of encryption, and QUIC in particular, traditional deep packet inspection methods are no longer effective for network traffic analysis, since the payload is no longer accessible. Recent research has therefore turned towards machine learning based approaches that rely only on flow level and packet level metadata, such as size, packet direction and timestamp, to identify the underlying application without decrypting the traffic itself. In this project, multi-branch deep learning framework is proposed and evaluated for classifying QUIC traffic generated by five Google Services, using publicly available UC Davis QUIC dataset of 6,439 labelled flows. Three sequence encoding architectures, Temporal Convolutional Network (TCN), 1D Residual Network, and CNN-BiGRU with Additive Attention were compared and best performing of these was combined with statistical and burst feature branch in fusion architecture. The final model achieved a macro F1-Score of 99.74 % on the held out test set, confirmed using five fold cross validation with a mean F1-Score of 99.54 % and a standard deviation of 0.20 %, indicating the result was not simply due to a favourable data split. An ablation study found each feature branch performs reasonably well individually, between 96% and 98.6% F1-Score, with fusion giving a modest but consistent improvement. This project also examines two aspects rarely tested directly in existing literature: real time classification feasibility and interpretability. Using only the first five packets of a flow, the model retains over 95% of its full flow performance, and Grad-CAM and attention analysis shows different applications exhibit distinct temporal patterns, some identifiable within the first few packets, others only later. These findings offer a more complete evaluation of encrypted traffic classification than accuracy alone.

I. INTRODUCTION

The increasing implementation of encryption protocols has significantly enhanced the privacy and security of Internet communications, but it has also created challenges for traditional network traffic analysis techniques. Traditional traffic classification methods, such as Deep Packet Inspection that rely on accessing packet payloads to identify applications and services. However, the prevalent use of an encrypted protocols has reduced the effectiveness of such approaches. Among these protocols, Quick UDP Internet Connections (QUIC) has gained significant attention due to its improved performance, reduced latency, and built in security features. As the underlying transport protocol for HTTP/3, QUIC is

expeditiously becoming a major component of modern internet communication and making its traffic analysis an important area of research.

To address the limitations of payload based inspection, researchers have increasingly explored machine learning techniques for encrypted traffic classification. These approaches utilize flow-level characteristics, packet size, and rest statistical features to identify traffic patterns without decrypting the actual packet content. Various machine learning algorithms, including Decision Trees, Random Forests, Support Vector Machines, and deep learning models, have demonstrated promising results in classifying QUIC traffic generated by different applications and services.

This literature review examines recent studies on machine learning-based QUIC traffic classification, focusing on feature extraction methods, classification techniques, performance evaluation, and the challenges associated with analyzing encrypted network traffic.

II. LITERATURE REVIEW

A. Methodology For Paper Selection

Relevant research papers were identified through a literature search using academic databases such as Google Scholar, IEEE Xplore, and ScienceDirect. Papers were selected based on their relevance to topic, their citation score and how recent they are. Priority was given to recent studies that reported classification methodologies, datasets, evaluation metrics, and key findings related to QUIC network traffic analysis.

B. Literature Review of Selected Studies

Tong et al. addressed the challenge of classifying encrypted QUIC traffic, which limits the effectiveness of traditional port-based and payload-based monitoring techniques [1]. They proposed a two-stage machine learning framework that combines flow-based and packet-based features. In the first stage, a Random Forest classifier utilizes eight statistical flow features to distinguish Google Hangout Chat and Voice Call traffic from other services. The remaining traffic is then classified into File Transfer, Video Streaming, or Google Play Music using a one-dimensional CNN that analyzes 1400 packet-based features extracted from the packets of each flow. The framework was evaluated on a custom 150 GB dataset containing over 20,000

flows collected over three weeks, with extensive preprocessing and 5-fold cross-validation. Experimental results showed that using 1400 features and a Sparse Categorical Crossentropy loss function achieved a micro-averaged F1-score of 99.24%. A major strength of the study is its hybrid feature extraction strategy, which enables accurate application-level classification without relying on decrypted payloads. However, the high-dimensional packet representation and complete flow-level feature extraction introduce significant computational overhead, potentially limiting scalability in high-volume network environments. The study demonstrates the effectiveness of combining statistical and deep learning approaches for encrypted traffic analysis and highlights the need for more computationally efficient architectures capable of maintaining high classification accuracy in large-scale network deployments.

Almuhammadi et al. addressed the challenge of QUIC encrypted traffic classification, noting that deep learning approaches often require large datasets and extensive parameter tuning, which limits their suitability for real-time deployment [2]. To provide a computationally efficient alternative, they evaluated five ensemble learning models, namely Random Forest, Extra Trees, Gradient Boosting Tree, Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LGBM), for application-level traffic classification. Using a public dataset from the University of California at Davis containing 5,939 encrypted flows across five Google services, the framework relied on lightweight early-stage packet features, including packet length, inter-arrival time, and packet direction, rather than payload decryption or full-flow statistics. Performance was assessed using accuracy, precision, recall, and F1-score under both train-test and 10-fold cross-validation settings. Experimental results showed that boosting-based models consistently outperformed bagging methods, with LGBM achieving 99.40% accuracy and F1-score using the first 120 packets, while still maintaining 92.31% accuracy with only 15 packets and a testing time of 0.017 seconds. A key strength of the study is its demonstration that lightweight ensemble models can achieve performance comparable to deep learning approaches while remaining computationally efficient and suitable for real-time deployment. However, the evaluation was limited to a small dataset containing only five Google-based applications, which may not adequately represent the diversity and complexity of real-world enterprise network traffic. The study highlights the potential of efficient ensemble learning techniques for encrypted traffic classification and emphasizes the need for validation on larger and more heterogeneous datasets.

Merlin et al. addressed the challenge of accurate QUIC traffic classification for network management, performance optimization, and security monitoring [3]. They proposed a machine learning framework based on ensemble learning and systematic hyperparameter optimization, evaluating five models: Random Forest, Extra Trees, Gradient Boosting Tree, XGBoost, and LightGBM. The framework was tested on the UC Davis QUIC dataset containing traffic from five Google services. Network flows were represented using the first 120

packets of each flow, with packet length, inter-arrival time, and packet direction extracted as features and normalized prior to training. To improve model performance and generalization, the authors employed Stratified K-Fold cross-validation and Randomized Search CV, while using early stopping for XGBoost and LightGBM to reduce overfitting. Experimental results demonstrated that hyperparameter tuning significantly improved classification performance, with the optimized Random Forest model achieving the highest accuracy of 91.9%, outperforming Extra Trees (90.6%) and XGBoost (88.0%). A key strength of the study is its detailed and practical hyperparameter optimization methodology, which provides valuable guidance for deploying ensemble learning models in encrypted traffic classification tasks. However, the evaluation was limited to traffic from five Google-based services, restricting the framework's ability to represent the diversity of real-world QUIC environments. Additionally, limiting analysis to the first 120 packets of a flow may exclude important characteristics of long-duration applications and multimedia streams. The study highlights the importance of model optimization in improving encrypted traffic classification performance while emphasizing the need for broader evaluation across more diverse traffic scenarios.

Bai and Tang addressed the challenge of fine-grained classification of encrypted QUIC traffic, where traditional machine learning approaches based solely on flow statistics suffer from reduced accuracy due to encryption-induced packet size distortion [4]. They proposed a two-stage framework in which a rule-based algorithm first identifies QUIC flows using protocol characteristics, ports, and packet padding patterns, after which the plaintext Server Name Indication (SNI) is extracted from the QUIC initial handshake and combined with ten statistical flow features for classification using CatBoost. Evaluated on the COMMON-APPS and CESNET-QUIC22 datasets, the framework achieved approximately 99.95% accuracy and a ROC-AUC of 0.9999, compared with about 73% accuracy when using statistical features alone. A key strength of the study is its practical and operationally viable hybrid design, which effectively integrates protocol inspection with machine learning and is suitable for deployment in real-world enterprise networks. However, the approach relies heavily on the availability of plaintext SNI, making it vulnerable to emerging privacy technologies such as Encrypted Client Hello (ECH), which may significantly reduce classification performance. The study highlights the importance of developing resilient multi-modal traffic classification techniques capable of maintaining high accuracy in future fully encrypted network environments, making it highly relevant to research on hybrid traffic classification and intelligent network management systems.

Gahtan et al. addressed the lack of comprehensive and standardized datasets for QUIC traffic analysis by introducing VisQUIC, a large-scale benchmark dataset containing over 100,000 labeled QUIC traces collected from more than 44,000 Tranco websites over four months [5]. The dataset was captured across two residential networks and uniquely in-

cludes paired SSL keys to enable controlled decryption and ground-truth validation, while covering multiple QUIC server implementations such as Chromium, mvfst, and quiche. In addition to providing packet captures, the authors proposed an image-based traffic representation that transforms packet length and temporal features into RGB images without requiring payload decryption. To demonstrate the dataset's utility, they developed a deep learning benchmark for estimating the number of multiplexed HTTP/3 responses within encrypted traffic. Traffic traces were converted into 32×32 images using sliding time windows, and the models were trained using data augmentation, a composite loss function, and standard optimization techniques. Experimental results showed strong predictive performance, achieving up to 97% accuracy within a margin of two responses and 92.6% cumulative accuracy across complete traces. A key strength of the study is the creation of a publicly available, large-scale benchmarking resource that supports reproducible research in QUIC traffic analysis while enabling both encrypted and controlled decrypted investigations. However, the dataset is limited to browser-generated web traffic and does not capture the diverse characteristics of mobile, WebRTC, or IoT environments. Additionally, performance declined when longer temporal windows were used, indicating challenges in maintaining accuracy over extended communication sessions. The study provides a valuable foundation for future research in encrypted traffic classification, protocol fingerprinting, and network security, while highlighting the need for broader traffic diversity and more robust temporal feature extraction methods.

Joarder and Fung addressed the growing threat of DDoS attacks against the QUIC protocol, particularly handshake flooding attacks that exploit the computational imbalance between clients and servers during connection establishment [6]. To improve upon existing defenses that suffer from high false positive rates and limited adaptability, they proposed QUICwand, a hybrid mitigation framework that combines a Dynamic Counting Bloom Filter with Bayesian Optimization and LightGBM. The framework dynamically adjusts Bloom Filter parameters, including filter size, hash functions, and traffic adjustment factors, to balance resource utilization and detection accuracy under changing network conditions. QUICwand was evaluated in a virtualized QUIC environment using the aioquic implementation and a hybrid dataset consisting of synthetic QUIC traffic and adapted DDoS attack patterns derived from the DARPA dataset. Performance was assessed using detection accuracy, false positive rate, anomaly detection scores, and resource utilization metrics. Experimental results demonstrated that QUICwand achieved detection rates exceeding 80% and reaching nearly 96% under certain attack scenarios, while maintaining lower false positive rates and reducing CPU, memory, and bandwidth consumption compared with baseline approaches. A key strength of the study is its integration of probabilistic data structures with machine learning-based optimization, enabling adaptive and resource-efficient attack mitigation in dynamic network environments. However, the evaluation relied heavily on a

synthetically adapted version of the DARPA dataset, which does not fully reflect the complexity, encryption characteristics, and multiplexed behavior of modern HTTP/3 and QUIC traffic. The study highlights the potential of adaptive machine learning-driven defense mechanisms for QUIC security while emphasizing the need for validation in large-scale, real-world environments and against more sophisticated application-layer DDoS attacks.

Kadi et al. addressed the challenge of detecting DDoS attacks within encrypted QUIC traffic, where the adoption of HTTP/3 limits the effectiveness of traditional payload-based monitoring techniques [7]. To overcome this limitation, they proposed a supervised machine learning framework that distinguishes legitimate HTTP/3 traffic from QUIC flood attacks using statistical flow characteristics. The study evaluated four classifiers, namely Support Vector Machine (SVM), Logistic Regression, Random Forest, and K-Nearest Neighbors (KNN), trained on 23 flow-based features including packet size distribution, inter-arrival time, data rate, and protocol-specific connection metrics. The framework was tested in a containerized HTTP/3 environment that generated both legitimate browsing traffic and attack traffic, producing over 15 GB of packet captures and 50,000 one-second traffic samples. Feature relevance was analyzed using t-SNE and ANOVA, which identified byte rate, packet rate, and packet size variability as the most discriminative attributes. Experimental results demonstrated excellent classification performance, with Random Forest and KNN achieving 100% accuracy, while SVM and Logistic Regression exceeded 86% accuracy across test scenarios. A key strength of the study is its practical demonstration that lightweight flow-based features can effectively detect QUIC-specific DDoS attacks without requiring access to encrypted payloads. However, the evaluation relied entirely on synthetically generated traffic within a controlled Docker environment and considered only HTTP GET flood attacks, limiting the assessment of model generalizability to real-world network conditions and more sophisticated attack types. The study highlights the effectiveness of machine learning for encrypted traffic anomaly detection and emphasizes the need for validation on diverse, real-world QUIC datasets and broader DDoS attack scenarios.

Kadi et al. addressed the challenge of detecting Layer 7 DDoS attacks in encrypted QUIC traffic, where transport-layer encryption limits the effectiveness of traditional intrusion detection systems [8]. To overcome this limitation, they proposed a machine learning framework that analyzes continuous bidirectional traffic flows using 64 statistical features derived from QUIC traffic characteristics. The study evaluated eight supervised learning algorithms, including Random Forest, Gradient Boosting, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP), for attack detection. Due to the lack of publicly available QUIC attack datasets, the framework was tested on a custom dataset of approximately 1 TB of traffic collected from a virtualized environment that simulated legitimate browsing alongside QUIC connection floods, GET floods, and HTTP scanning attacks. Feature engineering in-

corporated metrics such as packet inter-arrival times, packet direction entropy, and QUIC packet type distributions, while PCA and ANOVA were applied for dimensionality reduction and feature selection. Experimental results showed exceptional performance, with all models achieving ROC-AUC scores of 1.0 and F1-scores approaching 0.999. Random Forest provided the best balance between detection accuracy and computational efficiency, maintaining near-zero false positive rates even when the feature set was reduced substantially. A key strength of the study is its dynamic flow-based analysis approach and the creation of a large-scale QUIC-specific attack dataset tailored to modern HTTP/3 environments. However, the evaluation was conducted entirely within a simulated environment, and the near-perfect classification results suggest that the dataset may not fully reflect the complexity, noise, and feature overlap present in real-world networks. The study demonstrates the potential of transport-layer metadata for detecting attacks in encrypted environments while highlighting the need for validation on heterogeneous, real-world QUIC traffic and assessment of real-time deployment scalability.

Kadi et al. addressed the challenge of detecting DDoS attacks in encrypted QUIC and HTTP/3 environments, where transport-layer encryption reduces the effectiveness of traditional intrusion detection systems [9]. To identify malicious botnet traffic, they proposed a supervised machine learning framework evaluated within a virtualized HTTP/3 testbed. The experimental environment simulated realistic user browsing through Selenium WebDriver while a 40-node botnet generated application-layer GET flood attacks against an Nginx HTTP/3 server. From 10,000 bidirectional network flows, comprising 7,000 benign and 3,000 attack flows, the authors extracted 25 statistical features related to packet sizes, inter-arrival times, and packet type entropy. After feature standardization, the framework evaluated several classifiers, including a Multi-Layer Perceptron (MLP), Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Naive Bayes, Support Vector Machine (SVM), and Logistic Regression. Experimental results showed that the MLP achieved the best performance with 99% accuracy and an F1-score of 0.99, closely followed by Random Forest and Decision Tree models. A key strength of the study is its combination of realistic traffic emulation and automated flow-based feature extraction, demonstrating that encrypted QUIC traffic can be effectively analyzed without payload inspection. However, the evaluation relied on a simulated dataset focused solely on GET flood attacks, limiting the assessment of model performance under the diverse traffic patterns and attack scenarios encountered in real-world networks. The study highlights the potential of machine learning for QUIC-based DDoS detection and emphasizes the need for validation in live environments, adaptive learning mechanisms, and broader threat models encompassing more sophisticated attack types.

Gahtan et al. addressed the challenge of monitoring multiplexed HTTP/3 traffic in encrypted QUIC environments, where network devices cannot directly determine the number of concurrent responses within a connection [10]. To improve

traffic observability, they proposed DecQUIC, a deep learning framework that converts QUIC traffic traces into RGB images representing packet direction, timing, and packet length distributions. The framework combines Convolutional Neural Networks (CNNs), Gated Recurrent Units (GRUs), and a self-attention mechanism to capture spatial and temporal traffic patterns, while a custom loss function integrates Focal Loss, Distance-Based Loss, and Ordinal Regression Loss to address class imbalance and preserve the ordinal nature of response-count prediction. DecQUIC was evaluated on a large-scale dataset containing more than 100,000 QUIC traces collected from 44,000 websites and 18 web servers over four months. Traffic traces were processed using sliding windows and image-based representations, generating over seven million training samples. Experimental results demonstrated strong performance, achieving up to 97% accuracy on individual traffic windows and 92.6% accuracy in estimating the total number of responses within complete traces. A key strength of the study is its novel shift from conventional traffic classification toward fine-grained estimation of HTTP/3 connection characteristics, enabling deeper visibility into encrypted network behavior without payload inspection. However, performance declined when analyzing highly multiplexed connections and previously unseen servers, largely due to the scarcity of high-concurrency examples in the training data. The study demonstrates the potential of deep learning for advanced encrypted traffic analysis while highlighting the need for improved handling of rare traffic patterns, better generalization to unseen environments, and more efficient real-time processing techniques.

C. Key Findings

The reviewed literature shows that machine learning (ML), deep learning (DL), and hybrid approaches are highly effective for encrypted QUIC traffic analysis, often achieving classification accuracies above 90% and, in some cases, close to perfect results. Feature selection plays a critical role in performance, with hybrid feature-fusion methods significantly improving accuracy, though at the cost of increased computational overhead.

Studies also demonstrate that simple packet-level features, such as packet length, direction, and inter-arrival time, can achieve high accuracy when combined with ensemble learning algorithms like Random Forest, XGBoost, and LightGBM, making them suitable for real-time deployment, means that model can be used in live network environment to analyze traffic and make predictions immediately or with low latency. Deep learning models enhance analysis by uncovering complex spatial-temporal patterns, enabling accurate inference of application behaviors despite encryption.

In security applications, ML-based approaches have shown outstanding performance in QUIC attack and DDoS detection, with detection accuracy of more than 95% and low false-positive rates. But, challenges remain, including over dependence on large labeled datasets, rigorous feature engineering,

high computational requirements, and limited validation in real-world environments.

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D. Comparative Table

Tables I and II compare recent studies on QUIC traffic classification and security detection.

III. RESEARCH GAPS

Despite the high classification accuracies demonstrated in existing QUIC traffic analysis studies, most approaches rely heavily on handcrafted flow-level and packet-level features, extensive preprocessing, hyperparameter tuning, and fully supervised machine learning or deep learning models. These methods often require large labeled datasets and significant computational resources, which can limit scalability and increase deployment complexity. Furthermore, many studies have been evaluated using controlled or simulated datasets, raising concerns about their ability to generalize effectively to diverse real-world network environments.

Furthermore, current research has largely focused on improving classification performance rather than developing adaptive frameworks capable of automatically learning meaningful representations from encrypted traffic. The reliance on predefined features may restrict the ability of existing models to capture complex and underlying traffic patterns associated with new QUIC implementations and applications. Hence, there is also a need for more robust and scalable approaches that reduce dependence on manual feature engineering, enhance generalization to unseen traffic, and maintain high classification performance in dynamic encrypted network environments.

IV. PROPOSED METHODOLOGY

A. Feature Representation

Each flow is represented by three features derived solely from packet metadata i.e. timestamp, IAT, size and packet direction. Statistical compromises 13 flow-level attributes such as packet and byte counts, packet-size, flow duration, inter-arrival time statistics. Burst features are obtained by grouping consecutive packet same direction packets and 17 attributes describing burst-length, duration, and entropy of packets size and burst-length distributions. Sequence features represent each flow as an ordered series of packets. Flows are padded or truncated to fixed length of 60-packets so that they can be processed by sequence encoders. The statistical and burst features summarise the flow in aggregate and sequence features summarise the temporal order that aggregation discards.

B. Sequence Encoder Architectures

Three candidate architectures were implemented for encoding the sequence branch of the model. The first is Temporal Convolutional Network (TCN), which uses stacked 1D convolutions with increasing dilation to capture dependencies over long ranges within the packet sequence. The second one is ResNet1D, consisting of convolutional blocks with skip

connections while mitigating the vanishing gradients. The third is hybrid CNN-BiGRU architecture with additive attention, in which convolution layer extracts local features from packet sequence, the Bidirectional Gate Recurrent Unit then models the sequence in both temporal directions, and attention mechanism produces a weighted summary of sequence together with a set of per packet attention weights. All three were trained under identical data splits, optimiser settings, and stopping criteria, the architecture with the highest validation F1 was selected for further experiments.

C. Hyperparameter

Once the best architecture was chosen, a grid search was run to tune four hyperparameters of the CNN-BiGRU-Attention model: dropout (0.2 or 0.4), branch hidden size (32 or 64), batch size (32 or 64), and learning rate (0.001 or 0.0005). This gave sixteen combinations in total, each trained under the same conditions and compared on validation accuracy. The best configuration turned out to be dropout 0.2, hidden size 64, batch size 32, and a learning rate of 0.001, reaching 99.79% validation accuracy. These settings were then used throughout the rest of the study, in the ablation experiments, cross-validation, and early-classification tests.

D. Fusion Model

The classification model follows multi-branch design. The sequence branch uses the selected sequence model as described above, and statistical and burst features were passed through multi-layer perceptron to produce fixed length embeddings. The resulting embeddings are concatenated and passed through a fully connected classification head to produce the final class (Google Service) prediction. All models were trained using the cross entropy loss function and the Adam optimiser, with learning rate scheduler that reduces the learning rate when the validation loss plateaus, and early stopping based on validation loss to prevent overfitting.

E. Ablation Study

To evaluate the contribution of each feature branch, the fusion model was retrained using different combinations of branches, namely statistical features alone, sequence features alone, burst features alone, and all three branches combined. Each configuration used the same underlying architecture, hyperparameters, and training procedure, differing only in which branches were active, so that any difference in performance can be attributed to the availability of features than other factors. This isolates the effect of feature availability and quantifies whether fusion improves on any single branch used alone.

F. Early Classification

To evaluate the suitability for real-time deployment, classification was performed again using only the first K-packets of each flow, for K=5,10,15,20,30,45. Features were recomputed and rescaled from the truncated sequence at each K and the model was retrained from fixed initialisation to isolate genuine

TABLE I
SUMMARY OF QUIC TRAFFIC CLASSIFICATION AND DETECTION METHODOLOGIES

Author(s)	Title	Methodology	Dataset
Tong et al. (2018)	A Novel QUIC Traffic Classifier Based on CNNs	Two-stage Random Forest and 1D-CNN using flow-based and packet-based features.	QUIC traffic from five Google services
Almuhammadi et al. (2023)	QUIC Network Traffic Classification Using Ensemble Machine Learning Techniques	RF, ET, GBT, XGBoost, and LightGBM using packet length, direction, and inter-arrival time features.	UC Davis QUIC dataset
Merlin et al. (2024)	Hyperparameter Optimization for Ensemble Techniques in Classifying QUIC Traffic	RF, ET, GBT, XGBoost, and LightGBM with hyperparameter optimization.	UC Davis QUIC dataset
Bai & Tang (2024)	QUIC Traffic Classification Based on Multi-Feature Fusion	Rule matching, CatBoost, and multimodal CNN-MLP using SNI and flow features.	CESNET-QUIC22
Gahtan et al. (2024)	Exploring QUIC Dynamics: A Large-Scale Dataset for Encrypted Traffic Analysis	Image-based traffic representation with custom loss functions.	100,000+ QUIC traces from 44,000 websites
Joarder & Fung (2024)	QUICwand: A Machine Learning Optimization-based Hybrid Defense Approach Against QUIC Flooding Attacks	Dynamic Bloom Filter, LightGBM, and Bayesian Optimization.	Laboratory-generated QUIC attack dataset
Kadi et al. (2024)	Machine Learning for QUIC traffic flood detection	RF, KNN, SVM, and Logistic Regression with feature selection.	Simulated QUIC flood attack dataset
Kadi et al. (2025)	Adapting to the Evolution: Enhancing Intrusion Detection Through Machine Learning in the QUIC Protocol Era	Gradient Boosting, RF, and MLP with PCA and ANOVA feature reduction.	Simulated QUIC traffic
Kadi et al.(2025)	Refining Intrusion Detection in the Age of QUIC: Leveraging Machine Learning for Enhanced Security	Neural Network, RF, DT, KNN, SVM, and Logistic Regression.	Custom dataset of 10,000 flows
Gahtan et al. (2025)	Estimating the Number of HTTP/3 Responses in QUIC Using Deep Learning	CNN-GRU-Attention model using image-based representations of traffic.	100,000+ QUIC traces from 44,000 websites

TABLE II
PERFORMANCE EVALUATION AND KEY FINDINGS OF QUIC TRAFFIC ANALYSIS STUDIES

Author(s)	Results & Metrics	Key Findings	Strengths	Limitations
Tong et al. (2018)	F1-score = 99.24%	Flow and packet feature fusion improves classification accuracy.	Highly accurate QUIC classification.	High computational overhead.
Almuhammadi et al. (2023)	LightGBM achieved ~92% F1-score.	Simple packet features effectively classify encrypted traffic.	Suitable for real-time deployment.	Limited validation on diverse applications.
Merlin et al. (2024)	91.9% accuracy using optimized RF.	Hyperparameter tuning improves classification performance.	Interpretable ensemble models.	Sensitive to parameter selection.
Bai & Tang (2024)	99.95% accuracy.	SNI and flow feature fusion significantly boosts accuracy.	Extremely high classification performance.	Requires SNI extraction overhead.
Gahtan et al. (2024)	97% accuracy.	Image representations enable encrypted traffic analysis.	Large benchmark dataset and novel approach.	Computationally intensive.
Joarder & Fung (2024)	95.9% detection rate.	Dynamic optimization improves QUIC flood detection.	Adaptive and suitable for real-time mitigation.	Requires broader validation.
Kadi et al. (2024)	100% accuracy (RF and KNN).	ML effectively detects QUIC flood attacks.	Excellent attack detection performance.	Evaluated mainly on simulated traffic.
Kadi et al. (2025)	F1-score = 1.0, AUC = 1.0.	Feature optimization enhances intrusion detection.	Reduced feature complexity with high accuracy.	Potential overfitting to synthetic data.
Kadi et al. (2025)	99% Accuracy and F1-score.	Supervised ML effectively detects QUIC-based DDoS attacks.	Strong performance across attack scenarios.	Lack of real-world validation.
Gahtan et al. (2025)	97% accuracy (time windows); 92.6% (complete traces).	Deep learning infers HTTP/3 behavior from encrypted traffic.	Captures complex spatial-temporal patterns.	Requires large datasets and high computation.

information differences from training variance. Resulting performance was compared against the full-flow result to decide how many packets are needed to approach maximum accuracy.

G. Model Interpretability

Per-packet importance was computed for each correctly classified test flow. Grad-CAM applied to the final convolutional feature map for TCN and ResNet1D, and attention weights directly for CNN-BiGRU-Attention. Instead of looking at just one example, we averaged the scores of all correctly classified, this gave us more accurate overall all picture of what matter most.

H. Evaluation

Models were evaluated on a held-out test set using accuracy and macro-averaged precision, recall, and F1 score, with macro averaging weighting all classes equally regardless of size. 5-fold stratified cross-validation was carried out on the final fusion model to confirm results were not dependent on a single data split, with mean and standard deviation reported across folds.

V. DATASET USED

This study uses the UC Davis QUIC Traffic Dataset, a publicly available collection of encrypted network flow traces generated by five Google-hosted application services: Google Doc, Google Drive, Google Music, Google Search, and YouTube. Each flow represents a single client-server QUIC session captured at the packet level, recording only metadata observable without decryption, including timestamp, packet size, and direction, with all payload content excluded, consistent with the project's aim of classifying encrypted traffic without accessing packet contents. A total of 6,439 flows were loaded across the five classes, all parsed successfully after filtering out flows with fewer than five packets, which were deemed too short to carry reliable classification signal. The dataset is moderately imbalanced; stratified sampling was applied at every train/validation/test split and across all cross-validation folds to preserve this class distribution and avoid under representation of minority class during evaluation.

VI. RESULTS AND ANALYSIS

A. Architecture Comparison

The three candidate sequence encoders were trained and evaluated under identical conditions, using all three feature branches together. As shown in Table III, CNN-BiGRU with additive attention outperforms other two architectures i.e. TCN and ResNet1D, and was therefore selected as the sequence encoder for all subsequent experiments.

B. Ablation Study

Table IV summarises the performance of the fusion model under each feature-branch configuration. The fused model outperforms every individual branch, confirming that combining complementary feature views provides a measurable benefit. Among the individual branches, statistical features were the

TABLE III
PERFORMANCE COMPARISON OF SEQUENCE ENCODER ARCHITECTURES

Model	F1-Score	Inference Time (ms/sample)	Parameters	Size (MB)
CNN-BiGRU-Attn	0.998	9.254	21,590	0.082
ResNet1D	0.997	1.137	16,149	0.062
TCN	0.994	1.827	17,525	0.067

strongest predictor on their own, followed by sequence and then burst features. The fusion model improves the F1-score by approximately 1.1% points over the best single branch, but it also requires higher computational cost. This highlights a practical accuracy-versus-latency tradeoff that would need to be considered in any real deployment. As shown in Fig. 1,

TABLE IV
PERFORMANCE COMPARISON OF FEATURE CONFIGURATIONS

Configuration	F1-Score	Inference (ms/sample)
Statistical + Sequence + Burst	0.997	6.079
Statistical only	0.987	0.197
Sequence only	0.977	5.899
Burst only	0.961	0.197

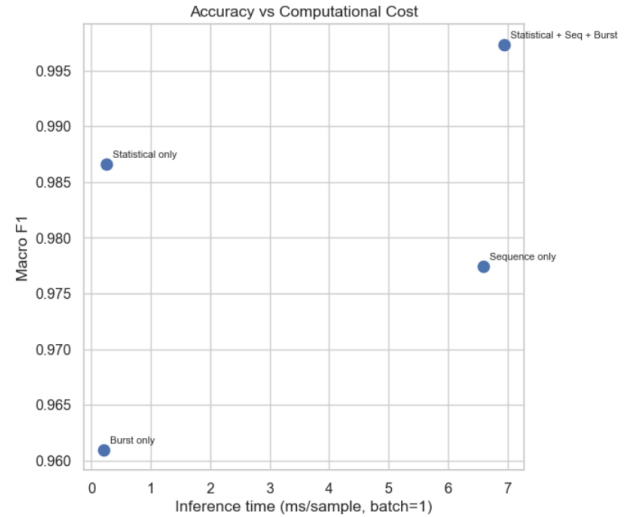


Fig. 1. Inference time versus F1-score for different feature-branch configurations.

the scatter plot of inference time versus F1-score highlights the accuracy-latency tradeoff, with Statistical-only and Burst-only configurations achieving lower inference times but lower F1-scores, while the Fusion and Sequence-only configurations provide higher F1-scores at a higher computational cost.

C. Cross-Validation

To confirm that the reported performance of the fused model was not dependent on a single train-test split, five-fold cross stratified cross-validation was performed. This produced a mean F1-score of 99.54% with standard deviation of 0.20% across all folds. This low variance indicates that the model's

performance is stable and reproducible across different train-test split, lending confidence to the single-split result reported in Table V.

TABLE V
FIVE-FOLD CROSS-VALIDATION RESULTS FOR THE FUSION MODEL

Fold	Accuracy	F1-Score
1	0.996	0.994
2	0.999	0.999
3	0.997	0.995
4	0.997	0.995
5	0.995	0.994

D. Model Interpretability

Class-averaged Grad-CAM (for the TCN and ResNet1D encoders) and attention weights (for the selected CNN-BiGRU-Attention encoder) were computed across all correctly classified test flows, to identify which packet positions in a flow most influence the model's predictions for each class. Different services rely on different parts of the flow. Some are identified from the first few packets, while others require later packets during payload transfer or sustained communication. As shown in Fig. 2, different services rely on different packet

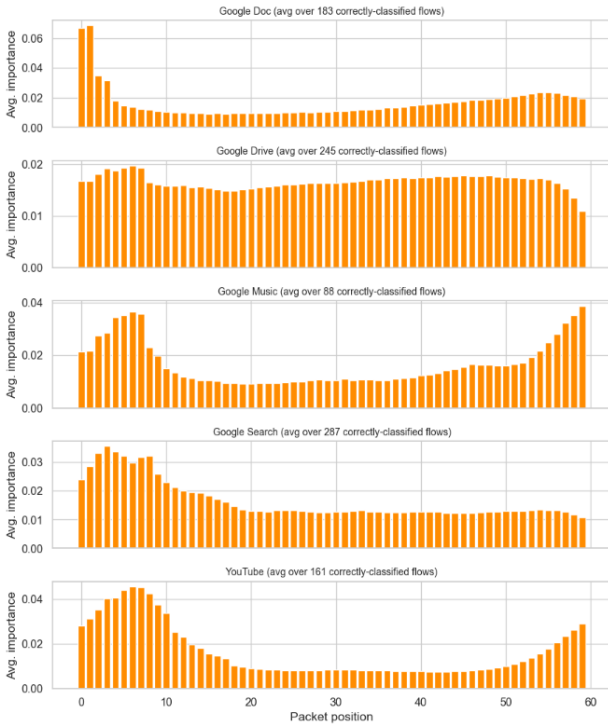


Fig. 2. Class-averaged feature importance across packet positions for different services.

positions for their distinguishing signals, supporting the early-classification analysis that follows.

E. Early Classification

Table VI shows model performance for different numbers of observed packets, from 5 to 45, compared with the 60-packet flow length used in this study.

TABLE VI
F1-SCORE FOR DIFFERENT NUMBERS OF OBSERVED PACKETS

K (packets observed)	F1-Score
5	0.962
10	0.979
15	0.980
20	0.983
30	0.974
45	0.984

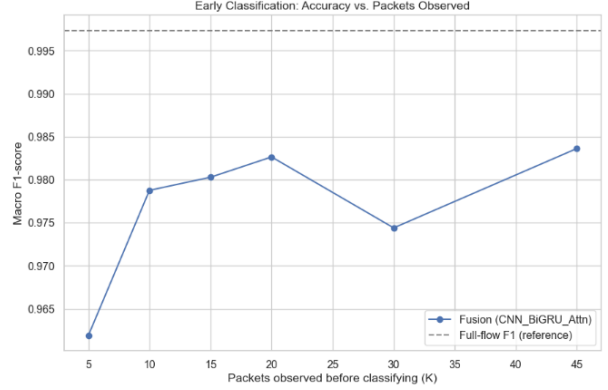


Fig. 3. Early classification performance as a function of the number of packets observed. The dashed line indicates the full-flow F1-score reference.

As shown in Fig. 3, the F1-score improves as more packets are observed, with strong performance achieved using only 10-20 packets. This supports the interpretability findings in Section D, where important class-specific signals were observed at different packet positions.

F. Summary

The results show that the fused model achieves the best overall performance i.e. F1-Score of 99.74%, cross-validated at $99.54\% \pm 0.20\%$, while the statistical-only model achieves a competitive F1-score of 98.66% with much lower inference time (0.20 ms/sample). Strong classification is also possible using only 10-20 packets, achieving F1-scores of 97.88%-98.26%. The interpretability results help explain this early-classification performance by showing where the important signals occur in the flow.

VII. CONCLUSION

This project developed and evaluated a multi branch deep learning framework for classifying encrypted QUIC traffic using only packet-level metadata. Fusing statistical, burst and sequence based features achieved macro F1-score of 99.74% on the held out test set confirmed via five fold cross validation where mean F1-score is 99.54% with standard deviation of 0.20%, indicating the result is stable than an artefact of single data split. Among the three sequence encoders compared, CNN-BiGRU with Additive attention performed best. The ablation study showed statistical features alone were the strongest individual predictor, ahead of sequence and burst features, with fusion improving the best single branch by roughly 1.1% points at the cost of approximately 27 times greater inference

latency, highlighting a clear accuracy vs efficiency tradeoff relevant to deployment choice. Beyond accuracy, this work tested two aspects largely assumed but rarely measured in prior literature, real time feasibility and interpretability. The model retained over 95% of its full flow performance using just five packets, plateauing bu 15 to 20, while GradCAM and attantion analysis showed distinguishing signal appear at different points in the flow. Limitations include evaluation on a single dataset from one source, with no cross-dataset or cross-time generalisation testing. This project demonstrates that high accuracy encrypted traffic classification is achievable using only metadata features, while providing practical insight into computational cost, early classification feasibility, and interpretability required for reliable real world deployment.

VIII. ACKNOWLEDGMENT

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IX. USE OF AI

AI tools like NotebookLM were used to support the understanding of research papers by providing high-level explanations of methodologies, frameworks, and approaches, along with simple examples to break down complex concepts. AI was also utilized for language refinement, including punctuation and grammar, as well as writing detailed Markdown documentation and debugging code. AI tools like Claude were also used to fix LaTeX rendering issues by correcting syntax in Overleaf.

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